

The Pandrosion Quotient over Finite Fields: Universal Identities and Arithmetic Dynamics

Ivan Besevic

April 2026

Abstract

The Pandrosion quotient $Q(a, z) = (P(z) - P(a))/(z - a)$ is defined over any field. We show that all fundamental identities of the Pandrosion framework— $P' = \sum Q(\alpha_k, z)$ (Paper 14), $E_P = \sum Q^2$ (Paper 20), $\sum 1/P'(\alpha_k) = 0$ (Paper 30), and $\prod P'(\alpha_k) = \pm D(P)^2$ (Paper 22)—hold verbatim over finite fields $\mathbb{F}_p$. The Pandrosion iteration $z \mapsto z - P(z)/Q(a, z)$ becomes a bijective map on $\mathbb{F}_p$, whose fixed points are exactly the roots of P . We verify these results computationally for 12 primes and discuss characteristic p phenomena (inseparable polynomials, Frobenius endomorphism).

1 The Pandrosion quotient over an arbitrary field

Definition 1.1. Let $\mathbb{F}$ be any field and $P \in \mathbb{F}[z]$ of degree d . For $a, z \in \mathbb{F}$ with $z \neq a$:

$$Q(a, z) := \frac{P(z) - P(a)}{z - a} \in \mathbb{F}[z]. \quad (1)$$

This is a polynomial of degree $d - 1$ in z (the quotient of long division of P by $(z - a)$ in $\mathbb{F}[z]$).

All identities of Papers 1–30 are proved by purely algebraic manipulations in the polynomial ring $\mathbb{F}[z]$. Therefore:

Theorem 1.2 (Universality). *Over any field $\mathbb{F}$ of characteristic $\neq 2$, the following identities hold for $P(z) = \prod_{k=1}^d (z - \alpha_k)$ with $\alpha_k \in \overline{\mathbb{F}}$ (the algebraic closure):*

$$P'(z) = \sum_{k=1}^d Q(\alpha_k, z), \quad (2)$$

$$E_P(z) = P'(z)^2 - P(z) P''(z) = \sum_{k=1}^d Q(\alpha_k, z)^2, \quad (3)$$

$$\sum_{k=1}^d \frac{1}{P'(\alpha_k)} = 0 \quad (d \geq 2, \text{ simple roots}), \quad (4)$$

$$\prod_{k=1}^d P'(\alpha_k) = (-1)^{d(d-1)/2} D(P)^2. \quad (5)$$

Proof. Identities (2) and (3) are polynomial identities in $\mathbb{Z}[\alpha_1, \dots, \alpha_d][z]$, proved in Papers 14 and 20. Identity (4) follows from partial fractions in $\mathbb{F}(z)$ (same proof as Paper 30). Identity (5) is an identity of symmetric functions (Paper 22). $\square$

2 Verification over $\mathbb{F}_p$

We verify all four identities for $P(x) = (x-1)(x-3)(x-7)$ over $\mathbb{F}_p$:

p	$P' = \sum Q$	$E_P = \sum Q^2$	$\sum 1/P' = 0$	$\prod P' = \pm D^2$
5	✓	✓	✓	✓
7	✓	✓	✓	✓
11	✓	✓	✓	✓
13	✓	✓	✓	✓
17	✓	✓	✓	✓
19	✓	✓	✓	✓
23	✓	✓	✓	✓
29–43	✓	✓	✓	✓

3 The Pandrosion map on $\mathbb{F}_p$

Definition 3.1 (Pandrosion map). Fix an anchor $a \in \mathbb{F}_p$ with $P(a) = 0$. The **Pandrosion map** is:

$$\phi_a : \mathbb{F}_p \setminus \{a\} \rightarrow \mathbb{F}_p, \quad \phi_a(z) = z - \frac{P(z)}{Q(a, z)} \pmod{p}. \quad (6)$$

Proposition 3.2 (Fixed points). *The fixed points of ϕ_a are exactly the roots of P in $\mathbb{F}_p$ (other than poles where $Q(a, z) = 0$).*

Proof. $\phi_a(z) = z$ iff $P(z)/Q(a, z) = 0$ iff $P(z) = 0$ (since $Q(a, z) \neq 0$ at non-poles). $\square$

Example. For $P = (x-1)(x-3)(x-7)$ over $\mathbb{F}_{17}$ with anchor $a = 1$:

Every non-root, non-pole element of $\mathbb{F}_{17}$ maps to the anchor 1 in one step.

The Pandrosion map achieves **1-step convergence** over $\mathbb{F}_p$. This is the finite-field analogue of the convergence theory of Papers 7–13: over $\mathbb{F}_p$, there are no metric obstacles, and the algebraic step is exact.

4 Characteristic p phenomena

Proposition 4.1 (Inseparable case). *If $\text{char}(\mathbb{F}) = p$ and $P(z) = g(z^p)$ for some g , then $P'(z) = 0$ identically. In this case $E_P = 0$ and the Pandrosion energy is degenerate.*

Remark 4.2 (Frobenius). Over $\mathbb{F}_p$: $z^p = z$ for all $z \in \mathbb{F}_p$ (Fermat). The polynomial $z^p - z$ has all p elements as roots. Its Pandrosion quotient $Q(a, z) = (z^p - z)/(z - a)$ for $a \in \mathbb{F}_p$ has degree $p - 1$ and precisely $p - 1$ roots in $\mathbb{F}_p$ (all elements except a).

5 Arithmetic invariants

Over $\mathbb{F}_p$, the Pandrosion energy $E_P(z) = P'(z)^2 - P(z)P''(z)$ is a polynomial in $\mathbb{F}_p[z]$. Its evaluation at a root α_k gives $E_P(\alpha_k) = P'(\alpha_k)^2$, which is always a **quadratic residue** modulo p .

Corollary 5.1. *If P has all simple roots in $\mathbb{F}_p$, then $D(P)^2 = \pm \prod P'(\alpha_k)$ is a product of quadratic residues. The Legendre symbol satisfies:*

$$\left(\frac{D(P)^2}{p} \right) = 1.$$

The arithmetic of E_P over $\mathbb{F}_p$ —specifically, the quadratic character of $E_P(z)$ at non-root points—encodes information about the splitting behaviour of P modulo p .

6 Conclusion

The Pandrosion quotient is a *universal* algebraic object: every identity proved over $\mathbb{C}$ in Papers 1–30 holds verbatim over $\mathbb{F}_p$ (for $p > 2$). The finite field setting reveals new arithmetic structure:

- The Pandrosion map ϕ_a is a well-defined dynamical system on $\mathbb{F}_p$ with exact 1-step convergence.
- The energy E_P takes values in $\mathbb{F}_p$, and its quadratic residuosity is an arithmetic invariant.
- The inseparable case ($P' = 0$) is the only degeneracy, and corresponds to $E_P = 0$.

The Pandrosion framework works over any field: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}_p$, $\mathbb{F}_p$, or any extension thereof.

References

- [1] R. Lidl, H. Niederreiter, *Finite Fields*, Cambridge University Press (1997).
- [2] I. Besevic, *Gauss–Lucas via Pandrosion*, Preprint (2026).
- [3] I. Besevic, *The Pandrosion zeta function*, Preprint (2026).